

DD_SUB-WF-PAUSE-01 - Subscription Pause

Developer implementation guide: DD_SUB-WF-PAUSE-01-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

SUB-WF-PAUSE-01 defines voluntary subscription pause.

Pause moves a subscription from:

ACTIVE -> SUSPENDED

It is used when the customer or an admin wants to temporarily stop service without terminating the subscription.

Examples:

- Customer vacation.
- Customer temporarily away.
- Financial hardship pause handled by call center.
- Admin maintenance pause.
- Admin compliance hold.

Pause is different from non-payment suspension:

PAUSE = voluntary/admin business pause
SUSPEND_NP = billing/dunning-driven non-payment suspension

Both may end in SUSPENDED, but they write different `origin_intent` values in `subscription_pause_history`.

2. Ownership

This DD owns:

- Pause trigger API: POST /api/subscriptions/{subscription_id}/pause.
- Pause reason catalog.
- Pause history table.
- Pause operator config.
- Pause duration model.
- Pause business rules.
- Pause domain events.
- Pause worker usage.

This DD does not own:

- WIK-specific BPMN. See DD_SUB-WF-PAUSE-01-FLOW-WIK.
- Subscription master state. See SUB-LM-01.
- Operation orchestration framework. See SUB-WF-FRAMEWORK-01.
- Worker topic contracts. See DD_SUB-WF-FRAMEWORK-01-WORKERS.
- Network suspension implementation. See FUL-04.
- Billing fee calculation. See BIL DDs.
- Resume operation. See SUB-WF-RESUME-01.
- Non-payment suspension. See SUB-WF-SUSPEND-NP-01.
- Notification template/channel delivery. See NOT-01.
- Scheduled resume scanner. See scheduler/foundation scheduling module.

3. Mental Model

Pause is a saga:

```
flowchart LR
    API[Pause endpoint] --> FW[SUB-WF framework]
    FW --> BPMN[Pause BPMN]
    BPMN --> RULES[Drools validation]
    BPMN --> LM1[SUB-LM PENDING_PAUSE]
    BPMN --> BIL[Billing pause intent/fee]
    BPMN --> FUL[FUL-04 suspend service]
    BPMN --> HIST[Pause history row]
    BPMN --> LM2[SUB-LM SUSPENDED]
    BPMN --> EVT[Kafka event]
    BPMN --> NOTIF[Notification]
```

The pause history row is the important business record. RESUME later reads it to decide which resume triggers are allowed.

4. Deployment Unit Clarification

Pause is not a standalone service.

It is deployed as part of the subscription workflow capability:

- Pause route/controller in SUB-WF API.
- Pause config/catalog/history migrations.
- Pause BPMN, such as sub-pause-wik.
- Pause Drools package, such as rules.subscription.pause.wik.
- Shared workers from the worker catalog.

5. Triggers

Trigger	Meaning	Allowed roles
CUSTOMER_REQUESTED_PAUSE	Customer requested pause through self-service, call center, or operator.	CUSTOMER_SELF_SERVICE, CALL_CENTER, SUBSCRIPTION_OPERATOR
ADMIN_PAUSE	Operator/admin pauses for maintenance, compliance, or operational reason.	SUBSCRIPTION_ADMIN, SUBSCRIPTION_OPERATOR

CUSTOMER_SELF_SERVICE also requires:

- subscription_pause_config.customer_self_service_enabled = true
- pause_reason.customer_self_service_eligible = true
- pause_reason.admin_only = false

6. Duration Model

Pause request must resolve to exactly one duration mode.

Mode	Request shape	Meaning
OPEN_ENDED	openEnded=true	No scheduled resume. Human/customer/admin resumes later.
FIXED_DAYS	durationDays=N	System computes resume_scheduled_at = now + N days.
SCHEDULED_RESUME_TIME	resumeScheduledAt=<timestamp>	Resume should be triggered at the supplied time.

Rules:

- Multiple duration fields are invalid.
- No duration defaults to open-ended only if the reason allows open-ended.
- Scheduled resume values must respect operator and reason limits.

7. Data Model

7.1 pause_reason

What this table is for:

pause_reason is the admin-managed catalog of allowed pause reasons. It controls who may use a reason, how long the pause may last, whether billing cycle accrual pauses, whether fulfillment suspension is required, and whether a pause fee may apply.

Use this table when:

- Validating pauseReasonCode.
- Deciding if customer self-service can use a reason.
- Deciding whether open-ended pause is allowed.
- Enforcing max duration.
- Building the pause history snapshot.
- Passing reason policy to billing and fulfillment.

Do not use this table for:

- Per-customer approval decisions.
- Resume state.
- Non-payment suspension reason details.
- Billing amount calculation.

```
CREATE TABLE pause_reason (  
  code TEXT PRIMARY KEY,  
  description TEXT NOT NULL,  
  customer_self_service_eligible BOOLEAN NOT NULL DEFAULT FALSE,  
  admin_only BOOLEAN NOT NULL DEFAULT FALSE,  
  default_fee_billable_event_code TEXT NULL,  
  max_duration_days INTEGER NULL,  
  open_ended_allowed BOOLEAN NOT NULL DEFAULT FALSE,  
  pauses_bill_cycle BOOLEAN NOT NULL DEFAULT TRUE,  
  suspends_fulfillment BOOLEAN NOT NULL DEFAULT TRUE,  
  is_active BOOLEAN NOT NULL DEFAULT TRUE,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  updated_by TEXT NOT NULL  
);
```

Column descriptions:

Column	Purpose
code	Stable reason code used in requests and history.
description	Human-readable reason explanation.
customer_self_service_eligible	Whether customer portal/app may use this reason.
admin_only	Whether only admin/operator roles may use this reason.
default_fee_billable_event_code	Optional BIL-CFG event code used when pause fee applies.
max_duration_days	Max allowed pause length for this reason. Null means no reason-level max.
open_ended_allowed	Whether open-ended pause is allowed for this reason.
pauses_bill_cycle	Whether billing cycle accrual should pause during suspension.
suspends_fulfillment	Whether network/service suspension should happen through FUL-04.
is_active	Whether reason can be used for new pauses.
created_at	Creation timestamp.
updated_at	Last update timestamp.
updated_by	User/service that last changed the row.

Sample data:

code	description	customer_self_service_eligible	admin_only	max_duration_days	open_ended_allowed	pauses_bill_cycle	suspends_fulfillment
CUSTOMER_VACATION	Customer travelling / vacation	true	false	90	false	true	true

code	description	customer_self_service_eligible	admin_only	max_duration_days	open_ended_allowed	pauses_bill_cycle	suspends_fulfillment
CUSTOMER_TEMPORARY_AWAY	Customer temporarily away from home	true	false	30	false	true	true
CUSTOMER_FINANCIAL_HARDSHIP	Customer financial hardship	false	false	60	false	true	true
ADMIN_MAINTENANCE	Operator maintenance window	false	true	NULL	true	false	true
ADMIN_COMPLIANCE_HOLD	Compliance or regulatory hold	false	true	NULL	true	false	true

Sample meaning:

- CUSTOMER_VACATION can be used by self-service and must have a scheduled/fixed duration up to 90 days.
- ADMIN_COMPLIANCE_HOLD is admin-only and may be open-ended.
- Admin reasons do not pause billing cycle in WIK seed policy.

7.2 subscription_pause_history

What this table is for:

subscription_pause_history records every suspension-like period that RESUME must understand. PAUSE writes voluntary/admin rows. SUSPEND-NP writes non-payment rows. RESUME closes rows.

Use this table when:

- Creating a pause record after pause succeeds.
- Checking whether a subscription already has an open pause/suspension row.
- Showing pause history to support.
- Letting RESUME decide whether a resume trigger is allowed.
- Letting scheduler find rows with resume_scheduled_at.

Do not use this table to replace subscription.status_code. Current status remains in SUB-LM.

```

CREATE TABLE subscription_pause_history (
  pause_id                TEXT PRIMARY KEY,
  operator_code           TEXT NOT NULL,
  subscription_id         TEXT NOT NULL REFERENCES subscription(subscription_id),
  customer_id            TEXT NOT NULL,
  origin_intent           TEXT NOT NULL,
  system_managed         BOOLEAN NOT NULL,
  pause_reason_code       TEXT NULL REFERENCES pause_reason(code),
  pause_reason_snapshot   JSONB NULL,
  duration_mode          TEXT NULL,
  duration_days          INTEGER NULL,
  resume_scheduled_at    TIMESTAMPTZ NULL,
  pause_fee_billable_event_id TEXT NULL,
  reconnection_fee_billable_event_id TEXT NULL,
  suspended_at           TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  actual_resume_at       TIMESTAMPTZ NULL,
  resume_reason_code      TEXT NULL,
  resume_actor_user_id   TEXT NULL,
  resume_actor_role      TEXT NULL,
  admin_force_resume     BOOLEAN NOT NULL DEFAULT FALSE,
  admin_force_resume_reason TEXT NULL,
  dunning_reason_code    TEXT NULL,
  dunning_cycle_reference TEXT NULL,
  dunning_escalation_level INTEGER NULL,
  outstanding_debt_amount NUMERIC(18,2) NULL,
  outstanding_debt_currency TEXT NULL,
  dunning_cycle_reference_at_resume TEXT NULL,
  pause_correlation_id   TEXT NOT NULL,
  pause_actor_user_id    TEXT NOT NULL,
  pause_actor_role       TEXT NOT NULL,
  notes                 TEXT NULL,

  CHECK (origin_intent IN ('CUSTOMER_REQUESTED_PAUSE', 'ADMIN_PAUSE', 'SUSPEND_NP')),
  CHECK ((origin_intent = 'SUSPEND_NP') = system_managed),
  CHECK (duration_mode IS NULL OR duration_mode IN ('OPEN_ENDED', 'FIXED_DAYS', 'SCHEDULED_RESUME_TIME'))

```

```

);

CREATE UNIQUE INDEX uq_open_pause_per_subscription
ON subscription_pause_history (subscription_id)
WHERE actual_resume_at IS NULL;

CREATE INDEX idx_pause_history_subscription
ON subscription_pause_history (subscription_id, suspended_at DESC);

CREATE INDEX idx_pause_history_open_scheduled
ON subscription_pause_history (operator_code, resume_scheduled_at)
WHERE actual_resume_at IS NULL AND resume_scheduled_at IS NOT NULL;

```

Column descriptions:

Column	Purpose
pause_id	Unique pause/suspension history ID.
operator_code	Operator scope.
subscription_id	Subscription being paused/suspended.
customer_id	Customer reference for reporting/support.
origin_intent	Why this suspension exists: voluntary pause, admin pause, or non-payment.
system_managed	True only for SUSPEND_NP.
pause_reason_code	Pause reason for voluntary/admin pauses. Null for SUSPEND_NP.
pause_reason_snapshot	Full reason policy copied at pause time.
duration_mode	Open-ended, fixed-days, scheduled-resume-time. Null for SUSPEND_NP.
duration_days	Number of days when fixed-days mode is used.
resume_scheduled_at	When scheduler should trigger resume, if set.
pause_fee_billable_event_id	BIL event ID for pause fee, if charged.
reconnection_fee_billable_event_id	BIL event ID for resume/reconnection fee, set by RESUME.
suspended_at	When suspension period started.
actual_resume_at	When RESUME closed the row. Null means open.
resume_reason_code	Resume reason set by RESUME.
resume_actor_user_id	Actor that resumed.
resume_actor_role	Role that resumed.
admin_force_resume	Whether admin override was used.
admin_force_resume_reason	Admin reason for forced resume.
dunning_* fields	Populated only by SUSPEND-NP.
pause_correlation_id	Operation ID that created the row.
pause_actor_user_id	Actor that initiated pause/suspension.
pause_actor_role	Actor role at pause time.
notes	Optional support note.

Sample data:

pause_id	subscription_id	origin_intent	pause_reason_code	duration_mode	resume_schedule_d_at	suspended_at	actual_resume_at
pause_01HZ_KA MAU_VAC	sub_01HZ_KAMAU_FIBER	CUSTOMER_REQUESTED_PAUSE	CUSTOMER_VACATION	FIXED_DAYS	2026-04-15T00:00:00Z	2026-03-15T10:00:00Z	2026-04-15T00:00:18Z
pause_01HZ_KA MAU_SUSPNP	sub_01HZ_KAMAU_FIBER	SUSPEND_NP	NULL	NULL	NULL	2026-06-15T02:00:18Z	2026-06-22T10:00:06Z
pause_01HZ_OT IENO_CMPL	sub_01HZ_OTIENO_FIBER	ADMIN_PAUSE	ADMIN_COMPLIANCE_HOLD	OPEN_ENDED	NULL	2026-05-22T14:00:00Z	NULL

Sample meaning:

- First row is a customer vacation pause that was later auto-resumed.
- Second row is a non-payment suspension written by SUSPEND-NP, not PAUSE.
- Third row is an open admin compliance hold.

7.3 subscription_pause_config

What this table is for:

subscription_pause_config stores operator-level switches and duration limits for pause.

Use this table when:

- Checking if self-service pause is enabled.
- Checking if scheduled/fixed-duration pause is enabled.
- Applying max future scheduled resume window.
- Applying minimum pause duration.
- Deciding whether to send customer notification.

```
CREATE TABLE subscription_pause_config (  
  operator_code          TEXT PRIMARY KEY,  
  customer_self_service_enabled  BOOLEAN NOT NULL DEFAULT FALSE,  
  scheduled_pause_enabled  BOOLEAN NOT NULL DEFAULT TRUE,  
  max_future_scheduled_resume_days  INTEGER NOT NULL DEFAULT 90,  
  min_pause_hours         INTEGER NOT NULL DEFAULT 24,  
  customer_notification_enabled  BOOLEAN NOT NULL DEFAULT TRUE,  
  updated_at             TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  updated_by            TEXT NOT NULL  
);
```

Column descriptions:

Column	Purpose
operator_code	Operator this config applies to.
customer_self_service_enabled	Master switch for customer self-service pause.
scheduled_pause_enabled	Allows FIXED_DAYS and SCHEDULED_RESUME_TIME.
max_future_scheduled_resume_days	Upper bound for scheduled resume time.
min_pause_hours	Prevents very short pauses.
customer_notification_enabled	Whether BPMN should call notification worker on success.
updated_at	Last update timestamp.
updated_by	User/service that updated row.

Sample data:

operator_code	customer_self_service_enabled	scheduled_pause_enabled	max_future_scheduled_resume_days	min_pause_hours	customer_notification_enabled
WIK	true	true	90	24	true
WUG	false	true	90	24	true
WTZ	false	true	90	24	true
YASSN	false	true	90	24	true

Sample meaning:

- WIK allows customer self-service pause.
- Other operators are conservative placeholders.
- All sample operators require at least 24 hours and allow scheduled pause up to 90 days.

8. Business Rules

Rule	Description
R-PAUSE-S-1	Pause requires subscription status ACTIVE.
R-PAUSE-S-2	Transition is ACTIVE -> PENDING_PAUSE -> SUSPENDED.
R-PAUSE-S-3	At most one open pause/suspension history row per subscription.
R-PAUSE-T-1	Trigger must be CUSTOMER_REQUESTED_PAUSE or ADMIN_PAUSE.
R-PAUSE-T-2	Trigger role must match allowed role set.

Rule	Description
R-PAUSE-T-3	Customer self-service requires operator config enabled.
R-PAUSE-T-4	Customer self-service requires reason eligibility.
R-PAUSE-R-1	Pause reason must be active.
R-PAUSE-R-2	Reason row must be snapshotted into history.
R-PAUSE-D-1	Exactly one duration mode must be resolved.
R-PAUSE-D-2	Open-ended requires reason support.
R-PAUSE-D-3	Duration must respect min/max bounds.
R-PAUSE-D-4	Fixed days computes resume_scheduled_at.
R-PAUSE-D-5	Scheduler/RESUME handles actual scheduled resume.
R-PAUSE-B-1	BPMN sends PAUSE_INTENT to billing.
R-PAUSE-B-2	If fee applies and pay-first is required, BPMN waits for payment.
R-PAUSE-F-1	FUL-04 is called with intent=SUSPEND_SERVICE when fulfillment suspension is required.
R-PAUSE-H-1	Successful pause inserts open subscription_pause_history row.
R-PAUSE-H-2	Active restrictions are cleared when subscription becomes SUSPENDED.

9. REST APIs

9.1 Trigger Pause

POST /api/subscriptions/{subscription_id}/pause HTTP/1.1
 Authorization: Bearer <jwt-or-service-token>
 Idempotency-Key: pause-sub_01HZ_KAMAU-vacation-20260315
 Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "pauseTrigger": "CUSTOMER_REQUESTED_PAUSE",
  "pauseReasonCode": "CUSTOMER_VACATION",
  "durationDays": 31,
  "notes": "Customer travelling to Mombasa for holiday season",
  "initiatingWorkflowKind": "BO_UI",
  "initiatingWorkflowRef": "cc-ticket-2026-03-15-3301"
}
```

Success:

202 Accepted

```
{
  "operationId": "subop_01HZ_KAMAU_PAUSE",
  "operatorCode": "WIK",
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "operationKind": "PAUSE",
  "pauseTrigger": "CUSTOMER_REQUESTED_PAUSE",
  "bpmnProcessKey": "sub-pause-wik",
  "currentState": "INITIATED",
  "finalState": null,
  "resolvedResumeScheduledAt": "2026-04-15T10:00:00Z"
}
```

Common errors:

HTTP	Code	Meaning
400	UNKNOWN_OR_INACTIVE_PAUSE_REASON	Reason does not exist or is inactive.
400	MULTIPLE_DURATION_MODES	More than one duration mode supplied.
400	DURATION_REQUIRED	Reason requires a duration but none was supplied.
400	OPEN_ENDED_NOT_ALLOWED_FOR_REASON	Open-ended requested for reason that disallows it.
400	DURATION_EXCEEDS_REASON_MAX	Requested pause exceeds reason max.
400	SCHEDULED_PAUSE_NOT_ENABLED	Operator disabled scheduled/fixed pause.
403	ROLE_NOT_AUTHORIZED_FOR_TRIGGER	Actor role cannot use trigger.
403	PAUSE_REASON_NOT_CUSTOMER_SELF_SERVICE	Self-service actor used non-self-service reason.
409	INVALID_STATE_FOR_PAUSE	Subscription is not ACTIVE.

HTTP	Code	Meaning
409	ALREADY_PAUSED	Open pause/suspension row already exists.
409	OPERATION_IN_PROGRESS	Another state-changing operation is in-flight.

9.2 Pause History

GET /api/subscriptions/{subscription_id}/pause-history HTTP/1.1
 Authorization: Bearer <jwt>

Response:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "pauses": [
    {
      "pauseId": "pause_01HZ_KAMAU_VAC",
      "originIntent": "CUSTOMER_REQUESTED_PAUSE",
      "pauseReasonCode": "CUSTOMER_VACATION",
      "durationMode": "FIXED_DAYS",
      "durationDays": 31,
      "suspendedAt": "2026-03-15T10:00:18Z",
      "resumeScheduledAt": "2026-04-15T10:00:00Z",
      "actualResumeAt": "2026-04-15T00:00:18Z",
      "resumeReasonCode": "SCHEDULED_AUTO_RESUME"
    }
  ]
}
```

10. Worker Usage In This Operation

The worker list is a logical BPMN task list, not a deployment list.

Worker topic	Why pause uses it
drools-decide	Validate status, reason, trigger, role, duration, fee applicability.
sub-lm-put-pending-status	Move to PENDING_PAUSE.
sub-lm-compute-cycle-window	Build billing context.
bil01-emit-intent	Ask billing to decide pause fee/cycle impact.
bil01-charge-pre-authorize	Pay-first pause fee if applicable.
bil01-confirm-charge	Confirm pause fee after payment.
bil01-revert-pending-charge	Compensation for fee payment timeout/failure.
ful-call-restrict	Call FUL-04 with SUSPEND_SERVICE.
sub-lm-put-active-restrictions	Clear active restrictions after suspension.
sub-lm-put-master-fields	Insert pause history row and update status timestamp.
sub-lm-commit-state	Commit subscription to SUSPENDED.
sub-lm-revert-to-prior	Revert to ACTIVE during compensation.
kafka-emit	Emit pause success/rejected event.
notification-send	Notify customer/admin, reliability-isolated.
subscription-operation-finalize	Mark operation terminal.

11. Kafka Events

11.1 SubscriptionPaused

Topic:

sophix.subscription.subscription.paused

Emitted after:

- FUL-04 suspension succeeds or is skipped by reason policy.
- Pause history row is inserted.
- SUB-LM commits status to SUSPENDED.

Sample:

```
{
  "eventType": "SubscriptionPaused",
  "aggregateId": "sub_01HZ_KAMAU_FIBER",
  "timestamp": "2026-03-15T10:00:18Z",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_KAMAU_FIBER",
    "customerId": "cust_01HZ_KAMAU",
    "operationId": "subop_01HZ_KAMAU_PAUSE",
    "pauseId": "pause_01HZ_KAMAU_VAC",
    "priorStatus": "ACTIVE",
    "newStatus": "SUSPENDED",
    "originIntent": "CUSTOMER_REQUESTED_PAUSE",
    "pauseReasonCode": "CUSTOMER_VACATION",
    "durationMode": "FIXED_DAYS",
    "durationDays": 31,
    "resumeScheduledAt": "2026-04-15T10:00:00Z",
    "pauseFeeBillableEventId": null,
    "clearedRestrictions": [],
    "initiatingActorUserId": "cc-agent-wik-007",
    "initiatingActorRole": "CALL_CENTER"
  }
}
```

11.2 SubscriptionPauseRejected

Topic:

sophix.subscription.subscription.pauserejected

Emitted after rejection or compensation failure path finalizes.

12. Drools Package

Package pattern:

rules.subscription.pause.<operator>

For WIK:

rules.subscription.pause.wik

Required decisions:

Decision	Purpose
validate-preconditions	Status active, no open pause, reason active, role/trigger allowed.
resolve-duration-mode	Resolves open-ended/fixed-days/scheduled-resume-time.
compute-pause-fee-applicability	Decides whether pay-first fee path is needed.

13. Auth

Rules:

- Resolve actor from token, not request body.
- Customer self-service requires operator config and reason eligibility.
- Admin reasons require admin/operator roles.
- Service-to-service callers must use service accounts.

14. Infrastructure Recommendations

Pause does not need separate infrastructure from SUB-WF.

Deploy:

Artifact	Where
Pause endpoint/controller	SUB-WF API pods.
Pause migrations	Application DB migration pipeline.
Pause BPMN files	Camunda deployment pipeline.

Artifact	Where
Pause Drools packages	Drools/KIE deployment pipeline.
Pause worker usage	Worker pods/apps from worker DD.

Monitor:

- Pause request count.
- Pause failure count.
- Payment timeout count.
- FUL-04 failure count.
- Open pauses with scheduled resume in the past.
- Operations stuck in PENDING_PAUSE.

15. Cross-DD Dependencies

DD/module	Relationship
SUB-WF-FRAMEWORK-01	Operation rows, idempotency, concurrency, Camunda start.
DD_SUB-WF-FRAMEWORK-01-WORKERS	Worker topic contracts.
DD_SUB-WF-PAUSE-01-FLOW-WIK	WIK BPMN implementation.
SUB-LM-01	Subscription state and active restrictions.
SUB-WF-RESUME-01	Reads/closes pause history rows.
SUB-WF-SUSPEND-NP-01	Also writes subscription_pause_history with SUSPEND_NP.
BIL-01	Pause intent and fee handling.
FUL-04	Network/service suspension.
NOT-01	Notifications.
Scheduler	Fires scheduled auto-resume.

16. Build Order

Detailed flow is in DD_SUB-WF-PAUSE-01-DEV_IMPL_GUIDE-v1.0.md.

Recommended order:

1. Add pause tables and indexes.
2. Seed pause reasons and config.
3. Add framework operation config for PAUSE.
4. Implement pause API and history API.
5. Implement Drools facts/decisions.
6. Deploy first BPMN, normally WIK.
7. Wire workers and message catches.
8. Add compensation paths.
9. Add scheduler integration for scheduled resume.

17. Test Checklist

- [] Pause rejects non-ACTIVE subscriptions.
- [] Pause rejects unknown/inactive reason.
- [] Customer self-service reason gating works.
- [] Admin-only reason gating works.
- [] Duration modes are mutually exclusive.

- [] Duration min/max bounds are enforced.
- [] Open pause uniqueness is enforced.
- [] Success inserts pause history row.
- [] Success commits subscription to SUSPENDED.
- [] Success clears active restrictions.
- [] Success emits SubscriptionPaused.
- [] Failure emits SubscriptionPauseRejected.
- [] Payment timeout compensation reverts to ACTIVE.
- [] Fulfillment failure compensation reverts to ACTIVE.